

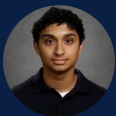
◆ **WDPR FALL 2026 INDUSTRIAL ENGINEERING CASE STUDY**

UC Berkeley Industrial Engineering and Operations Research

2026 CASE STUDY

More Magic per Minute

An attendance, capacity, and growth strategy for the
Autumn Harvest festival



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Case study objectives

**01**

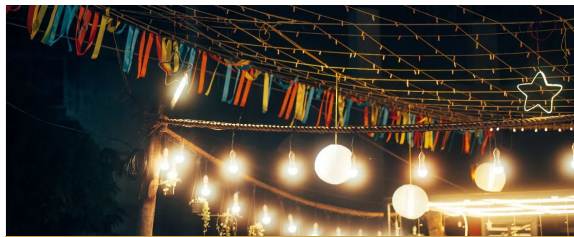
Forecast attendance

Estimate 2026 attendance by weekend, define a realistic planning range, and identify peak demand.

**02**

Recommend the vendor mix

Allocate the 120 booth-equivalent capacity across vendor types while balancing revenue and guest requirements.

**03**

Evaluate growth options

Compare expansion choices using financial impact, operational feasibility, and the guest experience.

Executive summary

Recommendations considering demand, vendor capacity, and growth around the same peak-day constraint.

PART 1

68,000

Estimating 68,000 visitors

Weekend 4 becomes the expected peak. The expected peak Saturday of 10,808 guests sets the design demand for the booth plan.

PART 2

45·21·27

Contracting 45 food, 21 merchandise, 27 activity booths

The mix fills all 120 space equivalents and keeps the longest expected peak wait near six minutes.

PART 3

12 hrs

Extending 10 to 12-hour days

Option A adds 20% more operating time, produces an expected \$55,000 annual benefit, and lowers expected peak food utilization to 76%.

OVERVIEW

Each part supplies the operating input required by the next decision.

1

Attendance demand

Estimate total visits, weekend demand, and the expected peak Saturday.

Output: 10,808 design-day guests

2

Booth capacity

Translate peak demand into food, merchandise, and activity capacity.

Output: 45 / 21 / 27 booths

3

Growth decision

Compare added hours and days against the remaining capacity gap and festival economics.

Output: extend to 12-hour days

01

PART 1

2026 Attendance Forecast

Total and weekend attendance

Part 1 | Forecast methodology, scenarios, and weekend demand



Part 1 problem and decision criteria

Festival leaders need an estimated attendance that reflects the changes unique to 2026.



QUESTION

How many visits should operations plan for?

An expected total, a planning range, and individual weekend-level demand with flexible assumptions are used for staffing and capacity decisions.

2026 CONSIDERATIONS

New Marketing Changes

Marketing reach expands, the festival marks its anniversary, and Weekend 4 includes a concert which affects the amount of visits.

DECISION OUTPUT

Demand by weekend

Find the expected peak weekend capacity to design plan for vendor capacity and variety for the best guest experience, operational feasibility and revenue generation.

2026 attendance forecast

68,000

expected visits

+21% vs. 2025

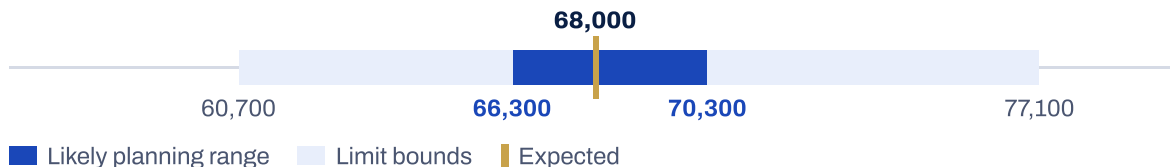
LIKELY PLANNING RANGE

66,300 to 70,300

Limit bounds: 60,700 to 77,100

Scenario	Visits	Growth	Visits vs. scenario high
Low	60,700	+8%	
Expected	68,000	+21%	
High	77,100	+38%	

PLANNING RANGE VS. STRESS LIMITS



NOTE The low and high cases are stress limits. The simulation range is the better planning range because drivers can offset one another.

Forecast methodology

We modeled growth based off prior year's data and the ten added factors present in 2026

1 Historical attendance (FP&A)

2021-2025 annual totals

2 Trend baseline

Average of five simple methods

3 2026 adjustments

Marketing, reach, anniversary, concert

4 Weekend allocation

Typical shares adjusted for the concert

FORECAST EQUATION

2026 attendance = baseline x marketing effect x anniversary effect + new-market visits + concert-only visits

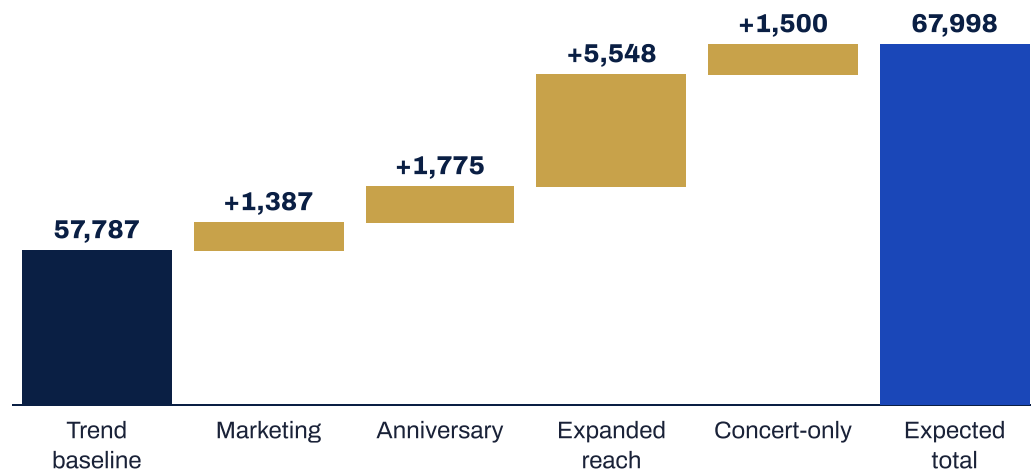
WHY 5 SIMPLE METHODS The lack of many annual observations cannot support a reliable seasonal or autoregressive model. Therefore taking the average of five simple methods is more representative, additionally the five included trend forecasts differ by only about 1,650 visits.

FIVE TREND FORECASTS FOR 2026



Sources of expected 2026 attendance growth

The changes for 2026 adds about 10,200 visits to the historical trend baseline.



Expected case, visits. Axis starts at 50,000.

57,800

trend baseline

+10,200

2026 driver uplift

LARGEST ASSUMPTION

How many residents in the newly reached metropolitan areas will attend?

Expected assumption: new markets convert at 8% of the local rate in year one.

EXPECTED TOTAL

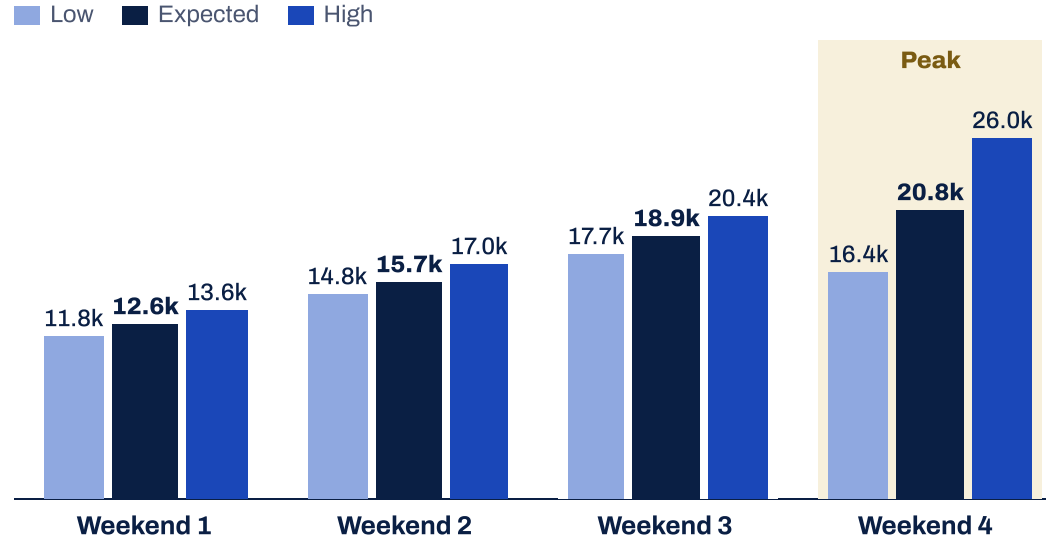
68,000 visits

TAKEAWAY

Approximately 15% of the expected forecast comes from 2026-specific adjustments rather than the trend.

Weekend attendance forecast

Attendance peaks at weekends 3 & 4



Weekend #	Low	Expected	High
1	11,800	12,600	13,600
2	14,800	15,700	17,000
3	17,700	18,900	20,400
4	16,400	20,800	26,000

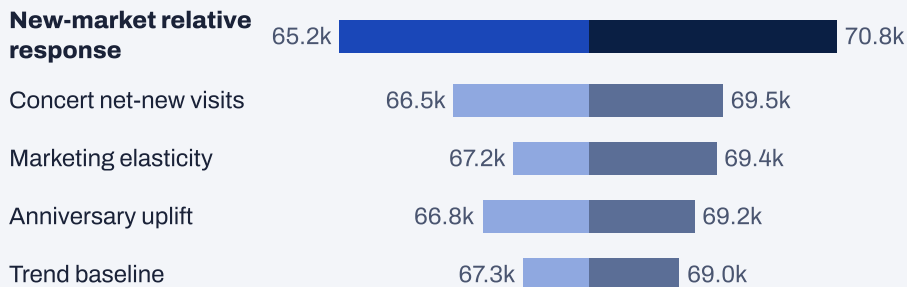


Forecasting risks and operating implications

New-market response drives forecast uncertainty. Limited concert capacity also creates operating constraints.

MAXIMUM MOVEMENT FROM EXPECTED FORECAST

Total visits when each driver moves alone from Low to High input



Expected 68.0k

EXPECTED WEEKEND 4 DEMAND VS. ASSUMED SEATS



4,800 potentially unseated guests

This is a stress test if every Weekend 4 visitor wants a seat. This assumes one 8,000-seat performance on Saturday and Sunday.



Part 1 takeaway

The forecast identifies the low, expected, and high attendance scenarios

RECOMMENDATION

68,000

**Plan for approximately
68,000 visits**

Use 66,300 to 70,300 as the likely planning range and retain the broader stress bounds for contingency planning.

OPERATING IMPLICATION

20,800

**Weekend 4 becomes
the expected peak**

Expected Weekend 4 attendance reaches approximately 20,800 visits, with Weekend 3 close enough to plan as a co-peak.

INPUT TO PART 2

10,808

10,808 peak-day guests

The expected peak Saturday becomes the design day for booth throughput and queue capacity.

MAIN UNCERTAINTY

The first-year response from residents in newly reached metropolitan areas.

02

PART 2

Vendor booth mix

Capacity first. Contribution second.



Part 2 problem and decision criteria

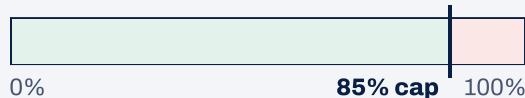
The booth plan must support peak demand without sacrificing the guest experience.



FIXED CAPACITY

120 space equivalents

Food and merchandise require one space per booth. Each activity booth requires two spaces.



SERVICE REQUIREMENT

Peak utilization no higher than 85%

The model first ensures that food booths can handle peak demand without excessive lines, then compares the financial results of the feasible options.



DECISION OUTPUT

Recommended booth mix

Select the combination of food, merchandise, and activity booths that maximizes vendor contribution while meeting all space, service, variety, and operating requirements.

Recommend 45 food, 21 merchandise, and 27 activity booths

The mix fills all 120 space equivalents and retains guest experience on the expected peak day.

FOOD & BEVERAGE

45 booths

45 spaces | 37.5%

MERCHANDISE

21 booths

21 spaces | 17.5%

ACTIVITIES

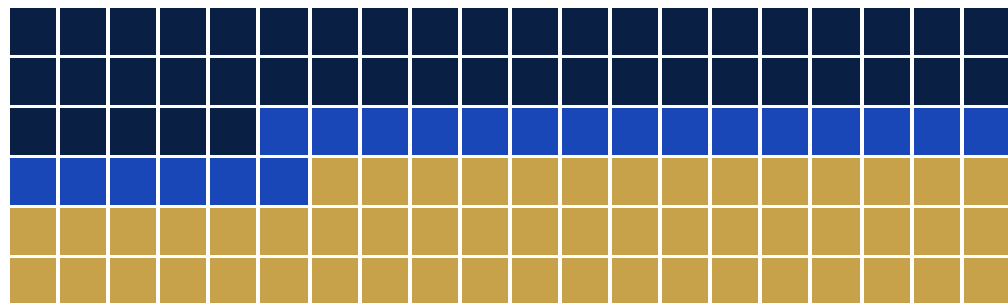
27 booths

54 spaces | 45.0%

EXPECTED PEAK DAY

\$394K gross vendor sales

\$156K contribution 10,808 guests



Each square = 1 of 120 space equivalents

■ Food ■ Merch ■ Activities

MAXIMUM PEAK WAIT

6.1 minutes

MAXIMUM UTILIZATION

82%

FEASIBLE MIXES TESTED

49 of 942

METHODOLOGY

Vendor mix must meet service, variety, and space constraints first. Then within that feasible set, the mix with the maximized profit is selected.

Queues are optimized; then the final 19 spaces are allocated

Part 1's 10,808-guest peak day is converted into peak-hour transactions, then into minimum booth counts.

Category	Daily txns	Peak-hour arrivals	Service / booth / hr	Min. booths @ 85%	Space equiv.
Food & Beverage	12,564	1,885	55	41	41
Merchandise	3,864	464	40	14	14
Activities & Experiences	5,188	674	35	23	46
Queueing minimum				78 booths	101 spaces

CAPACITY FORMULA

Minimum booths = $\text{CEILING}(\text{peak-hour arrivals} / (\text{service rate} \times 85\%))$

Example: Food = $\text{CEILING}(1,885 / (55 \times 0.85)) = 41$ booths

120 spaces

19

■ OPTIMIZER

$120 - 101 =$

**19
spaces**

allocated by the
optimizer

101

■ QUEUEING
MINIMUM

Why other vendor combinations do not work

The recommended mix earns more than the other alternative while keeping every category inside the service limit.

Mix	F / M / A	Max utilization	Longest wait	Gross sales	Service level
Recommended optimizer	45 / 21 / 27	82%	6.1 min	\$394K	PASS
Food-heavy	60 / 36 / 12	138%	Unbounded	\$338K	FAIL
Equal space	40 / 40 / 20	96%	45.2 min	\$353K	FAIL
Match participation	52 / 22 / 23	87%	11.9 min	\$374K	FAIL
Queueing minimums only	41 / 14 / 23	87%	11.9 min	\$352K	FAIL

With only 12 activity booths, utilization reaches 138%. Guests balk, activity sales disappear, and shorter stays reduce food transactions. This leads to the mix earning \$56K less in daily gross sales than the recommendation.

MAXIMUM UTILIZATION BY MIX



Stress testing the selected vendor mix

The same 45 / 21 / 27 mix is stress-tested against the Low, Expected, and High peak days from Part 1.

Design day	Guests	Food util.	Merch util.	Activity util.	Max wait	Service level
Low	9,218	70%	45%	67%	3.4 min	PASS
Expected	10,808	82%	52%	78%	6.1 min	PASS
High	13,498	102%	65%	98%	Unbounded	FAIL

HIGH-DAY
QUEUEING
MINIMUM

127
spaces

AVAILABLE
FOOTPRINT

120
spaces

STRUCTURAL
SHORTFALL

7 spaces

High-day queueing minimum

127

Available footprint

120



TAKEAWAY

Part 2 takeaway

RECOMMENDATION

45-21-27

**45 food, 21 merchandise,
27 activities**

The 93 physical booths consume all 120 space equivalents because activities require double space.

EXPECTED RESULT

\$394K

\$394,000 gross vendor sales

The longest expected peak wait is approximately 6.1 minutes and maximum utilization remains near 82%.

INPUT TO PART 3

7 spaces

Seven-space High-day shortfall

A 13,498-guest Saturday requires 127 spaces at the target utilization, so a booth re-mix cannot close the gap.

NEXT

Part 3 should test options that spread peak demand or add flexible service capacity.

03

PART 3

Growth recommendation

Compare the options and close the capacity gap.

Financial impact, operating feasibility, and guest experience



Part 3 problem and decision criteria

The festival is looking for ways to maximize growth by extending the festival duration in various ways.

DECISION

Choose the next growth move

Compare two more hours per day, four Fridays, and a fifth weekend using the same demand and cost framework.

KNOWN CONSTRAINT

Food service binds on a High Saturday

The Part 2 mix reaches 102% peak utilization in the High case. Growth should ideally spread demand away from the lunch peak.

DECISION STANDARD

Return, relief, and feasibility

Evaluate net-new demand, five-year value, congestion relief, vendor economics, and reversibility.

WHERE EACH OPTION ADDS OPERATING TIME

	Weekend 1		Weekend 2		Weekend 3		Weekend 4		Weekend 5	
Current	Sat	Sun	Sat	Sun	Sat	Sun	Sat	Sun		
A: 12-hour days	Sat +2h	Sun +2h	Sat +2h	Sun +2h	Sat +2h	Sun +2h	Sat +2h	Sun +2h		
B: add Fridays	Fri	Sat	Sun	Fri	Sat	Sun	Fri	Sat	Sun	
C: fifth weekend	Sat	Sun	Sat	Sun	Sat	Sun	Sat	Sun	Sat	Sun

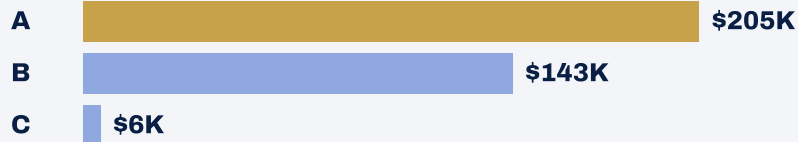
KEY DISTINCTION

Longer days spread demand within the same day. Fridays and a fifth weekend help only when guests move away from Saturday.

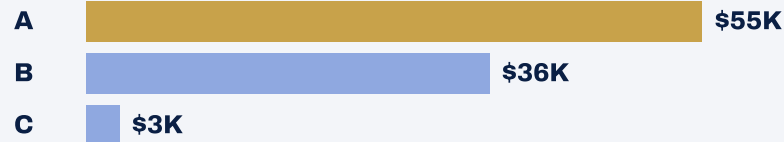
Part 3 option comparison

Option	Net-new visits	Year-1 benefit	5-year NPV	Expected / High food util.	Main feasibility issue
A: 12-hour days	5,440	\$55,000	\$205,000	76% / 95%	Confirm vendors can staff dinner hours
B: add Fridays	9,547	\$36,000	\$143,000	74% / 92%	Food booths lose about \$38 per Friday
C: fifth weekend	7,140	\$3,000	\$6,000	78% / 97%	Late-season weather and weak cushion

5-YEAR NPV



YEAR-1 BENEFIT



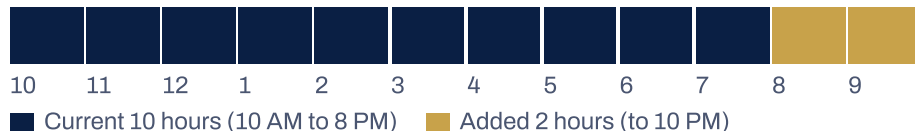
TAKEAWAY

Option A remains profitable in the Low case. Options B and C lose money when demand is weaker than expected.

Part 3 recommendation

RECOMMENDED OPTION

Extend each day from 10 to 12 hours



DECISION RATIONALE

Option A delivers the best expected five-year value, remains profitable in every scenario, and directly spreads the food-service peak across a longer day.



FINANCIAL RESULT

\$205,000 NPV

\$55,000 expected year-1 benefit with a 2.4x break-even cushion.

OPERATING RESULT

76% peak use

Expected food utilization falls from 82% to 76%. High demand remains at 95%.



IMPLEMENTATION SAFEGUARD

Confirm dinner-hour staffing with major food vendors and stage standby food trucks on Weekends 3 and 4.

Part 3 takeaway

The growth plan prioritizes the cheapest, reversible option while saving other scalable options for future dates, minimizing risk.


2027

Extend to 12-hour days

Open from 10 AM to 10 PM. Track hourly attendance, ZIP code based attendance, evening spending, and peak food utilization.


2028 or later

Hold the fifth weekend in reserve

Add Option C only if peak weekends remain above the 85% utilization target after longer days.


Not recommended

Do not add full-day Fridays

At expected traffic, a typical food booth loses about \$38 per Friday. Revisit only as an evening format.

DECISION TRIGGER

Decision trigger for Option C: evening hours carry at least 8% of visits and peak weekends still exceed 85% food utilization.

Final recommendation



DEMAND PLAN

Approximately 68,000 visits

Plan Weekend 4 as the expected peak and retain the Part 1 range for contingency staffing.



VENDOR PLAN

45 food, 21 merchandise, 27 activities

Contract the recommended mix for the expected peak day and validate food throughput before final commitments.



GROWTH PLAN

Extend to 12-hour days in 2027

Keep the fifth weekend as the next growth step. Do not add full-day Fridays under the current vendor economics.

CONTINGENCY

Pair longer days with standby food trucks on the busiest weekends because Option A alone leaves High-day food utilization at 95%.

Implementation priorities and next steps

1 Before vendor contracting

ACTION

Confirm dinner-hour staffing and expected evening sales with major food vendors

OWNER

Vendor and operations team

2 Before schedule lock

ACTION

Plan for additional lighting, staffing, and standby food trucks for high demand weekends (Weekends 3 and 4)

OWNER

Festival leadership

3 Before ticket release

ACTION

Publish the 10 AM to 10 PM schedule and set the 85% food-utilization trigger

OWNER

Guest experience team

4 During the 2027 festival

ACTION

Capture hourly gate counts, ZIP codes, evening spend, queues, and food throughput

OWNER

Analytics team

REVIEW TRIGGER

Option C review trigger criterion: evening demand reaches beyond 8% and peak food utilization remains above 85%

APPENDIX

Appendix

Supporting analysis

Detailed methods, assumptions, sensitivity, and references

A = attendance forecast B = vendor booth mix C = growth recommendation

A-C

APPENDIX · PART 1

Attendance forecast

Methodology, formulas, assumptions, and research

Appendix numbering: A = Part 1 B = Part 2 C = Part 3



Forecast method and data flow

The model separates the historical trend from the changes unique to 2026.

1

Historical baseline

2021-2025 attendance
Five simple trend methods

2

Accuracy check

Forecast 2024 and 2025
Measure MAPE and bias

3

2026 adjustments

Marketing, new markets,
anniversary, concert

4

Weekend allocation

Reweight 20 / 25 / 30 / 25
Add concert visits to W4

FORECAST EQUATION

**2026 visits = baseline x (1 + spend change x ad elasticity) x (1 + anniversary)
+ new-market visits + concert-only visits**

WHY AN ENSEMBLE Five annual observations cannot support a reliable complex time-series model. Averaging simple methods reduces dependence on any one trend assumption.

Historical baseline and ensemble calculation

Each method extends the same five annual observations in a different way.

Method	Formula used for 2026	Forecast	Included
Linear regression	$a + b \times 6$	57,100	Yes
Log-linear trend	$\exp(a + b \times 6)$	57,262	Yes
CAGR 2021-2025	$56,000 \times (56,000 / 49,000)^{(1/4)}$	57,901	Yes
CAGR 2022-2025	$56,000 \times (56,000 / 48,500)^{(1/3)}$	58,749	Yes
Average annual growth	$56,000 \times (1 + \text{average}[-1.0\%, 3.1\%, 7.0\%, 4.7\%])$	57,924	Yes
Equal-weight ensemble	Average of the five included forecasts	57,787	Selected

ACCURACY CHECK

Rolling hold-outs forecast 2024 and 2025 from earlier years. The included methods produced 3.6%-4.9% MAPE, and every method forecast below actual attendance.

METHOD SPREAD

1,649 visits



Expected-case attendance build

The expected forecast adds 10,210 visits to the history-based baseline.

$$57,787 \times 1.024 \times 1.03 + 5,548 + 1,500 = 67,998 \text{ visits}$$

Build step	Value	Increment	Basis
Historical baseline	57,787	-	Model
Marketing uplift	59,174	+1,387	Research
Anniversary halo	60,950	+1,775	Team
Expanded reach	66,498	+5,548	Mixed
Concert-only visits	67,998	+1,500	Team

MARKETING

20% spend increase x 0.12 elasticity = 2.4% uplift

$$57,787 \times 2.4\% = 1,387 \text{ visits}$$

EXPANDED REACH

New population = 1,200,000 - 500,000 = 700,000

Local penetration = (56,000 / 1.3) / 500,000 = 8.615%

$$\text{Reach visits} = 700,000 \times 8.615\% \times 8\% \times 1.15 = 5,548$$

NOTE Anniversary and concert effects have no direct published festival elasticity. The model labels them as team assumptions and tests low and high values.

Weekend allocation and concert capacity

The concert shifts four percentage points into Weekend 4, then the model adds 1,500 concert-only visits.

POOL BEFORE CONCERT-ONLY VISITS

60,950 existing-audience visits + 5,548 reach visits = 66,498

REWEIGHTING FACTOR FOR WEEKENDS 1-3

$(1 - 29\%) / (1 - 25\%) = 94.67\%$

Weekend	Base share	Adjusted share	Attendance calculation	Expected visits
1	20.00%	18.93%	66,498 x 18.93%	12,590
2	25.00%	23.67%	66,498 x 23.67%	15,738
3	30.00%	28.40%	66,498 x 28.40%	18,885
4	25.00%	29.00%	66,498 x 29.00% + 1,500	20,784
Total	100.00%	100.00%		67,998

CONCERT CAPACITY

8,000 seats x 2 performances = 16,000 seats

W4 visits  20,784

Seats  16,000

POTENTIAL SHORTFALL

20,784 - 16,000 =

4,784

Scenario and simulation methodology

Stress scenarios stack extreme assumptions. The simulation lets independent drivers offset one another.

Driver	Low	Expected	High
Trend baseline	57,100	57,787	58,749
Advertising elasticity	0.05	0.12	0.24
New-market conversion vs. local	4%	8%	12%
Anniversary halo	1%	3%	5%
Concert-only visits	0	1,500	3,000

SIMULATION LOGIC

1,000 trials sample each driver from a triangular distribution defined by its Low, Expected, and High values. The new-market response matters most: changing only that factor moves the expected forecast from about 65,200 to 70,800.

DISCRETE SCENARIOS

Low	60,712
Expected	67,998
High	77,056

MONTE CARLO PLANNING RANGE

P10	approximately 66,300
P50	approximately 68,300
P90	approximately 70,300

P10
|
~66.3k

P50
|
~68.3k

P90
|
~70.3k

Assumption register

Case facts, published evidence, and team judgments remain separate throughout the model.

Input	Expected	Type	Evidence
Marketing spend increase	20%	Case fact	Case [1]
Advertising reach	1.2M	Case fact	Case [1]
Advertising elasticity	0.12	Research	Sethuraman [5]; Henningsen [6]
Visits per local guest	1.3	Team	Converts visits to unique guests
New-market conversion	8% of local	Mixed	Distance evidence [7]; team rate
Anniversary halo	3%	Team	No published festival elasticity
Concert visits and shift	1,500; 4 pts	Team	Stress-tested Low to High
Concert performances	2	Team	One show per festival day
Case fact Case fact Research Research Team Team judgment Mixed Research concept plus team-selected rate			

Part 1 references

The numbered citations used throughout the appendix are listed below.

- [1] WDPR Fall 2026 IE Case Study. Historical attendance, weekend shares, marketing spend, advertising reach, anniversary concert, and venue capacity.
- [2] Makridakis, S. & Hibon, M. (2000). The M3-Competition: Results, conclusions and implications. *International Journal of Forecasting*, 16(4), 451-476.
- [3] Makridakis, S., Spiliotis, E. & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1), 54-74. ideas.repec.org/a/eee/intfor/v36y2020i1p54-74.html
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- [5] Sethuraman, R., Tellis, G. J. & Briesch, R. A. (2011). How Well Does Advertising Work? *Journal of Marketing Research*, 48(3), 457-471. papers.ssrn.com/ab_id=1866002
- [6] Henningsen, S., Heuke, R. & Clement, M. (2011). Determinants of advertising effectiveness. *Business Research*, 4(2), 193-239. doi.org/10.1007/BF03342755
- [7] Popp, N., Jensen, J. & Jackson, R. (2017). Maximizing visitors at college football bowl games. *International Journal of Event and Festival Management*, 8(3). doi.org/10.1108/IJEFM-02-2017-0014
- [8] Bass, F. M. (1969). A new product growth model for consumer durables. *Management Science*, 15(5), 215-227.
- [9] Sultan, F., Farley, J. U. & Lehmann, D. R. (1990). A meta-analysis of applications of diffusion models. *Journal of Marketing Research*, 27(1), 70-77.
- [10] Hyndman, R. J. & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679-688.
- [11] Hyndman, R. J. & Athanasopoulos, G. *Forecasting: Principles and Practice*, 3rd ed., Chapter 8. otexts.com/fpp3/

APPENDIX · PART 2

Vendor booth mix

Methodology, formulas, assumptions, sensitivity, and research

Appendix numbering: A = Part 1 B = Part 2 C = Part 3

A large, bold, yellow letter 'B' graphic, which is part of the appendix numbering system where B represents Part 2.

Three-layer decision model and data flow

Service and variety are constraints; contribution is maximized only after those conditions are met.

1 Demand + queues

10,808 guests → transactions → peak-hour arrivals → minimum booths

2 Integer optimization

Evaluate all 942 mixes; keep 49 feasible; maximize vendor contribution

3 Scenario stress test

Re-run the selected mix on 9,218 / 10,808 / 13,498 guests

LEXICOGRAPHIC DECISION RULE

1 Keep utilization $\leq 85\%$

2 Preserve category variety

3 Use ≤ 120 spaces

4 Maximize contribution

WHY ENUMERATION Three integer decisions and one space equation produce fewer than 1,000 candidate mixes, so every possibility can be audited.

SEARCH FUNNEL

942

candidate mixes

49

feasible

1

selected: 45 / 21 / 27

Demand, utilization, and wait-time formulas

The model turns attendance into capacity with explicit, editable assumptions for participation, purchasing, peaks, and service speed.

DAILY TRANSACTIONS

Guests x participation x transactions per participant

Food: $10,808 \times 77.5\% \times 1.5 = 12,564$

PEAK ARRIVALS (LAMBDA)

Daily transactions x peak-hour share

Food: $12,564 \times 15\% = 1,885$ per hour

UTILIZATION (RHO)

Peak arrivals / (booths x service rate)

Food: $1,885 / (45 \times 55) = 76\%$ before dwell; 82% with dwell

MINIMUM BOOTHS

CEILING($\lambda / (\text{service rate} \times 85\%)$)

Food: $\text{CEILING}(1,885 / (55 \times 85\%)) = 41$

M/M/1 QUEUE WAIT

$W_q = \rho / [\mu \times (1 - \rho)]$

At the recommended mix: 4.8 / 1.7 / 6.1 minutes

FLOW

**Guests → transactions → peak arrivals
→ utilization → booths → wait**

Optimization objective, behavioral links, and constraints

The case facts remain fixed; unsupported operating details are isolated as team assumptions and tested separately.

OBJECTIVE

Maximize total contribution = $\text{sum}[\text{captured transactions} \times \text{spend} \times (1 - \text{operating cost } \%)]$

CASE FACT

Space equation

$$F + M + 2A = 120$$

Activities use two standard spaces

RESEARCH + JUDGMENT

Operating cost

65% / 55% / 40%

Turns gross sales into contribution

QUEUE STANDARD

Peak utilization cap

85%

Keeps single-server waits manageable

TEAM JUDGMENT

Behavioral links

Dwell +1% | assortment 0.3 | balking 5-20 min

Captures complementary food demand, vendor variety, and lost sales from long lines

TEAM JUDGMENT

Variety bounds

**Merch $\geq 15\%$ | Activity 20-45%
| Food $\leq 50\%$**

Preserves festival identity; the 45% activity-space ceiling binds

Sensitivity, decision risk, and validation priorities

The result is robust to economics; throughput and the activity-space ceiling are the assumptions that can move the recommendation.

Assumption tested	Result	Interpretation	Impact
Operating margins +/- 5-10 points	Booth mix unchanged	Economics change dollars, not capacity	■ ■ ■ ■
Food service rate +/- 5 txns/hour	Moves 4-5 booths across food / merchandise	Largest practical throughput lever	■ ■ ■ ■
Dwell effect set to zero	42 / 24 / 27	Only three booths move	■ ■ ■ ■
Activity cap raised to 50%	30 activity booths; +\$1.7K/day	Current 45% cap binds	■ ■ ■ ■

MOST VALUABLE NEXT DATA POINT

Typical food orders cleared per booth during the lunch peak

IMPORTANT LIMITATION

Hourly averages hide short surges; the 85% cap is the buffer. The High day remains infeasible and moves to Part 3.

Part 2 references

Case facts, external benchmarks, and modeling references used in the booth-mix analysis.

- [1] Mobile Cuisine. Food truck food and labor cost percentages; prime-cost target below 60% and food cost 25-38%. mobile-cuisine.com/?p=23268; mobile-cuisine.com/?p=65149
- [2] Food & Hotel Asia (2025). Prime costs should range from 55% to 65% of total sales in an F&B business. foodnhotelasia.com/?p=12505
- [3] Priceva. Keystone pricing: retail price at roughly twice wholesale cost, implying a 50% gross margin. priceva.com/glossary/keystone-pricing
- [4] Craftybase. Pricing handmade products for wholesale; keystone pricing is a standard boutique and gift-retail expectation. craftybase.com/blog/pricing-handmade-products-wholesale
- [5] Little, J. D. C. (1961). A proof for the queuing formula $L = \lambda W$. *Operations Research*, 9(3), 383-387.
- [6] Charnes, A. & Cooper, W. W. (1961). *Management Models and Industrial Applications of Linear Programming*. Wiley. Goal programming and lexicographic objectives.
- [7] Hillier, F. S. & Lieberman, G. J. *Introduction to Operations Research*. Queueing theory (M/M/1, M/M/s) and integer programming.
- [8] Walt Disney Parks and Resorts. Associate Industrial Engineer job description: simulation, forecasting, optimization, queueing, and capacity/demand analysis. disneycareers.com

MODEL FILES booth_mix_model.xlsx; booth_mix_model_selection.pdf; booth_mix_model_selection_simplified.pdf. Part 1 inputs: festival_forecast_model.xlsx.

APPENDIX · PART 3

Growth options

Methodology, formulas, assumptions, sensitivity, and research

Appendix numbering: A = Part 1 B = Part 2 C = Part 3



Growth-option method and data flow

The model compares every option with the same demand, cost, and risk sequence.

1

Baseline

68,000 season visits and the Part 2 food-service constraint

2

Net-new demand

Estimate gross visits, then remove guests shifted from existing days

3

Incremental value

Count only added revenue, added operating cost, and one-time cost

4

Decision

Compare NPV, break-even cushion, peak relief, and feasibility

YEAR-1 NET BENEFIT

Net-new visits x \$13.97 + commission on added dwell sales - incremental operating cost

NPV Five-year NPV discounts the annual cash flows at 8% after the one-time option cost.

Option assumptions and formulas

The case defines the options. The model makes the missing demand and cost assumptions explicit.

Input	Low	Expected	High	How the model uses it
Option A evening uplift	4%	8%	12%	Season visits x uplift. No day-to-day cannibalization.
Option A existing-guest dwell uplift	1.5%	3.0%	4.5%	Adds commission revenue from guests staying through dinner.
Option B Friday share of Saturday	35%	45%	60%	Four Fridays x average Saturday visits x demand ratio.
Option B cannibalization	50%	40%	30%	Removes Friday guests who would have visited on the weekend.
Option C fifth-weekend size	60%	75%	90%	Scales a shoulder weekend, then removes cannibalized visits.
Option C weather impairment probability	35%	25%	15%	Revenue factor = 1 - probability x 50% attendance loss.

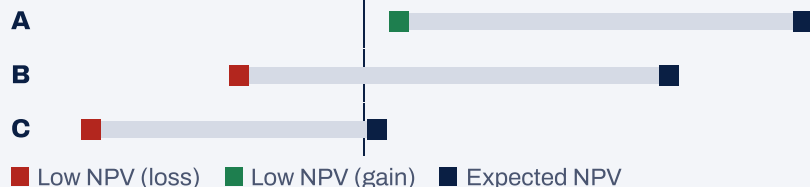
COMMON INPUTS \$10 admission yield, 15% vendor-sales commission, \$1.50 variable cost per visit, 8% discount rate, and five-year horizon.

Financial and operating comparison

Option A combines positive downside value with direct relief of the food-service peak.

Option	Low benefit	Expected benefit	High benefit	Low NPV	Expected NPV	Break-even cushion	Exp. / High food use
A: 12-hour days	\$12k	\$55k	\$99k	\$17k	\$205k	2.4x	76% / 95%
B: add Fridays	-\$11k	\$36k	\$110k	-\$58k	\$143k	1.4x	74% / 92%
C: fifth weekend	-\$28k	\$3k	\$43k	-\$127k	\$6k	1.0x	78% / 97%

LOW TO EXPECTED NPV RANGE



FRIDAY VENDOR CHECK

At 45% of Saturday traffic, a typical food booth earns about \$2,273 in sales and loses about \$38 after food cost and day-rate labor.

Sensitivity and limitations

Option A ranks first under every decision-weight set tested.

Weight set	A	B	C	Top
Balanced	4.35	2.75	2.45	A
Finance-heavy	4.20	2.85	2.40	A
Growth-heavy	3.90	3.15	3.30	A
Guest-experience-heavy	4.60	2.70	2.30	A
Risk-averse	4.55	2.50	2.30	A

MATERIAL LIMITATIONS

- 1 Demand response is judgmental. Option A protects the recommendation with a 2.4x expected break-even cushion.
- 2 The \$10 gate yield and \$27,000 daily cost are assumptions. Leadership should replace both with actuals.
- 3 Peak spreading and vendor willingness to staff two more hours require validation with hourly data and vendor outreach.
- 4 No option brings the High Saturday below the 85% food-utilization target without standby capacity.

Part 3 references

Model inputs, operating benchmarks, and decision-method references

- [1] Cal Poly Pomona Pumpkin Festival. 2024 Artisan Vendor Terms. Vendor fee equals 15% of gross sales.
- [2] Cal Poly Pomona Pumpkin Festival. 2023 Food Vendor Application. \$250 fee plus 15% of gross sales.
- [3] Anacortes Arts Festival. 2025 Food Vendor Terms. Commission equals 10% of sales.
- [4] Lebanon Strawberry Festival. 2025 Food Court Vendor Terms. Base fee or 20% of sales, whichever is greater.
- [5] State Fair of Texas. Daily Attendance, 2024. Friday attendance ranged from 56% to 86% of the following Saturday.
- [6] WRAL. North Carolina State Fair Daily Attendance, 2019. Fridays were approximately 69% to 70% of Saturdays.
- [7] Horngren, Datar, and Rajan. Cost Accounting: A Managerial Emphasis. Relevant-cost and incremental-decision methods.
- [8] Brealey, Myers, and Allen. Principles of Corporate Finance. NPV, IRR, payback, and real-options methods.

CASE AND MODEL FILES WDPR Fall 2026 IE Case Study, growth_options_model.xlsx, and growth_options_model_selection.pdf